

SMART INSOLE

START PRESENTATION

Project Title:

Smart Insole For Foot Ulcer Prediction

Supervisors:

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TEAM MEMBERS:

Hassan Sherif Hassan : Data Analysis, AI

Heba Muhamed Aly : Data Analysis, AI

Ahmed Sami Elsayed : Embedded System

Heba Ehab Muhamed : UI\UX Design

Ahmed Farag Khoudary : Mobile Application

INTRODUCTION

- **Diabetes can cause serious foot complications due to poor blood circulation and nerve damage.**
- **Most patients are unaware of early signs until the situation becomes critical.**
- **Our solution provides continuous foot monitoring using smart sensors to detect potential risks early.**
- **It also includes a Healthcare Page to check overall body health .**
- **he system also integrates deep learning to analyze uploaded foot images from the mobile app, helping detect visual signs of ulcers or infections early.**

Problem Definition

- **Diabetic patients suffer from foot ulcers, infections, or pressure injuries that may lead to amputation.**
- **Current monitoring methods are manual and not continuous.**
- **Patients need a smart, affordable, and reliable solution that can detect early warning signs automatically.**

Project Objectives

- 1. Develop a smart shoe that measures foot pressure, temperature, and humidity.**
- 2. Send real-time data to Firebase via ESP32.**
- 3. Use AI models to detect abnormalities and predict possible foot risks.**
- 4. Integrate a deep learning model to analyze foot images and classify them as normal or abnormal.(Shift the numbering down by one.)**
- 5. Build a mobile application that shows alerts and live readings.**
- 6. Add a Healthcare Page for users to manually enter vital data (BP, HR, sugar, etc.) and get an AI-based health evaluation.**

Tools & Technologies Used

Hardware:

- **ESP32 Microcontroller**
- **Pressure, Temperature & Humidity Sensors**
- **Rechargeable Battery**
- **Custom Insole Prototype**

Software & Platforms:

- **Arduino IDE (for ESP32 Programming)**
- **Firebase (Database & Cloud)**
- **Python (AI & Data Analysis)**
- **Scikit-Learn (Machine Learning)**
- **TensorFlow / Keras (Deep Learning – Image Classification)**
- **Flutter (Mobile Application)**
- **Figma (UI/UX Design)**
- **MS Project (Time Plan)**

TimeLine

Week 1-2

Week 3-5

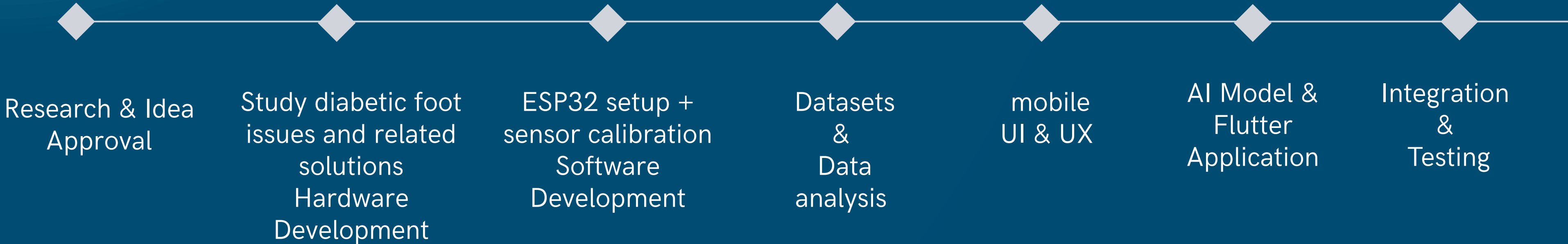
Week 6-9

Week 10-13

Week 14-17

Week 18-23

Week 24-28



Thank You